

HABORO, Japan

WMO: 474040

Lat: 44.3633N

Lon: 141.7008E

Elev: 31

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.4	10.9	0.7	5.7	9.4	4.2	6.8	13.3	27.6	26.3	25.7	25.3	6.2	120	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	80.4	71.7	77.9	70.4	75.9	69.4	73.7	78.2	72.3	76.4	70.8	74.8	8.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.3	119.7	76.5	70.8	113.7	75.4	69.2	107.4	74.0	37.4	78.4	36.1	76.1	34.8	75.1	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.7	22.6	20.1	-1.5	86.2	3.4	3.6	-3.9	88.7	-5.9	90.8	-7.7	92.9	-10.2	95.5	
			DB												
			WB	-2.1	75.7	3.3	2.2	-4.4	77.2	-6.4	78.5	-8.2	79.7	-10.6	81.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	23.2	24.0	31.1	41.4	52.1	59.6	67.2	69.9	63.1	51.8	39.6	28.7
	DBStd	17.21	5.81	6.65	6.12	5.92	6.22	5.24	4.56	4.41	5.57	5.70	6.90	5.72
	HDD50	3489	832	728	587	266	48	2	0	0	0	47	320	659
	HDD65	7213	1297	1148	1052	707	403	173	28	10	100	408	763	1124
	CDD50	2068	0	0	0	10	112	291	535	616	393	104	7	0
	CDD65	315	0	0	0	0	2	12	98	160	43	0	0	0
	CDH74	935	0	0	0	0	11	29	208	554	133	0	0	0
	CDH80	100	0	0	0	0	0	1	24	58	17	0	0	0
Wind	WSAvg	9.2	10.7	10.4	10.1	9.1	8.1	6.5	6.1	6.7	7.9	10.4	12.0	12.2
Precipitation	PrecAvg	50.9	4.4	3.2	2.5	2.3	2.8	2.6	5.0	4.8	5.8	5.7	6.1	5.6
	PrecMax	66.0	7.3	5.2	4.6	5.2	4.7	5.1	11.5	9.6	13.5	10.1	9.8	8.9
	PrecMin	38.9	2.7	1.6	0.6	0.6	0.9	1.0	0.7	1.4	1.9	2.8	2.9	3.1
	PrecStd	7.1	1.1	1.1	0.9	1.1	1.0	1.0	2.9	2.0	2.9	2.1	1.8	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.8	42.3	49.9	65.7	75.4	77.8	82.9	85.7	82.2	69.2	60.7	47.1
		MCWB	36.0	36.9	41.3	51.6	58.4	64.6	73.0	74.7	72.4	60.3	53.5	42.9
	2%	DB	36.0	38.8	45.6	59.4	69.3	73.1	78.9	81.0	77.3	66.5	56.5	43.0
		MCWB	32.5	34.9	39.8	49.2	56.7	63.2	70.8	72.8	69.5	59.1	50.9	39.2
	5%	DB	33.9	36.6	42.8	55.1	64.8	70.0	76.0	78.9	74.7	64.1	53.6	39.8
		MCWB	30.9	32.9	38.1	46.2	55.2	62.1	69.7	71.6	67.8	56.9	48.6	35.8
	10%	DB	31.9	34.2	40.3	51.1	61.6	67.5	74.0	76.9	72.4	61.8	51.0	37.3
		MCWB	28.9	30.9	36.2	44.8	54.1	61.3	68.8	70.6	65.8	55.2	45.9	33.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	38.2	43.4	53.8	61.4	68.7	74.7	75.9	73.9	63.3	54.6	43.8
		MCDB	38.9	41.4	48.2	61.9	68.7	73.5	79.7	81.8	79.6	67.2	59.6	46.9
	2%	WB	33.1	35.3	40.6	50.0	58.5	65.3	72.8	74.3	71.3	60.2	51.7	39.5
		MCDB	35.4	38.5	44.3	57.6	65.7	70.3	76.4	79.4	76.3	64.8	55.9	42.5
	5%	WB	31.2	33.2	38.5	47.5	56.6	63.5	71.1	73.0	68.9	58.1	49.0	36.3
		MCDB	33.6	36.2	42.1	53.5	63.1	68.1	74.5	77.1	73.5	63.1	53.2	39.2
	10%	WB	29.2	31.1	36.5	45.4	54.9	62.0	69.6	71.8	66.8	55.9	46.2	33.8
		MCDB	31.6	34.0	40.0	50.5	60.7	66.4	72.9	75.9	71.6	60.9	50.7	36.9
Mean Daily Temperature Range	5% DB	MDBR	11.3	12.3	11.8	13.4	13.9	11.6	10.5	11.7	14.6	13.6	10.4	9.2
		MCDBR	11.4	12.2	14.2	19.7	19.8	15.6	12.6	13.1	14.8	15.5	14.2	11.3
		MCWBR	9.9	10.5	10.5	11.5	10.5	8.2	6.7	7.2	8.9	10.3	11.7	10.0
	5% WB	MCDBR	11.1	11.8	13.1	17.6	16.8	12.5	10.5	11.1	12.9	14.1	14.0	11.2
		MCWBR	10.1	10.6	10.4	11.3	9.7	7.3	6.1	6.6	8.3	10.4	12.3	10.5
Clear-Sky Solar Irradiance	taub	0.349	0.400	0.458	0.519	0.533	0.516	0.492	0.460	0.410	0.413	0.395	0.369	
	taud	2.035	1.898	1.827	1.808	1.872	2.002	2.141	2.251	2.333	2.160	2.101	2.001	
	Ebn at Noon	228	236	239	237	239	244	249	253	257	233	211	203	
	Edh at Noon	34	47	58	64	62	55	47	41	35	36	32	32	
All-Sky Solar Radiation	RadAvg	343	612	1007	1464	1636	1616	1508	1433	1195	775	397	261	
	RadStd	36	81	101	133	139	183	185	102	104	60	52	25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page